COMPARATIVE BEHAVIOR OF PRIMATES[1]

II. DELAYED REACTION TESTS ON PRIMATES AT BRONX PARK ZOO

A. H. MASLOW AND H. F. HARLOW

University of Wisconsin

I. INTRODUCTION

This experiment was carried on as an extension of the work on the comparison of primates done at the Vilas Park Zoo in Madison, Wisconsin, by Harlow, Uehling and Maslow (2). The purpose of that investigation, as of this, was to compare the ability of primates (ranging from lemur to anthropoid) on tests of delayed reaction. The technique used in the present experiment was modeled as nearly as possible on that used in the previous study. Such differences as may be present were necessitated in every case by inflexible conditions and are, in the opinion of the author, of relatively little importance. They will be indicated in the appropriate place.

As regards subjects, there seems to be one important variation, namely, that the animals at the Bronx Zoo were of a more variable temper than the animals at Vilas Park. In many cases, the Bronx animals were ferocious and dangerous, so much so that some of them were unavailable for experimentation. These included 3 adult chimpanzees, and 3 adult orang-outans. Several other animals, notably a siamang gibbon, were unavailable because of their extreme timidity.

All of the animals (except the gorilla and chimpanzee) were housed in the primate house. This building was divided into two groups of cages: on one side of the house were lodged the small

[1] We wish here to express our thanks for and appreciation of the coöperation shown by Dr. Blair, director of the Bronx Zoological Gardens, and Dr. Charles V. Noback, veterinary surgeon at the Zoo. This experiment would have been impossible without their aid.

animals, many of them in each cage. These cages were extremely large and roomy, but there was no connection to the outside, so that they were confined indoors all the year round. It was impossible to experiment on the animals in these cages, both because of the number of animals in each cage and because of the close wire mesh which prevented the animals from reaching for food. Through the kindness of Dr. Noback, several cases at the hospital building were put at our disposal. It was thus possible to isolate several of the small animals and experiment on them under improved conditions. On the other side of the house were individual cages for the larger primates. Each cage was connected through a doorway to an outside cage. The floor was about 40 inches above the floor of the primate house, and was of concrete. In each cage there were wooden shelves for sleep and rest.

These cages were cleaned daily. The animals were fed a very light morning meal at about 8:00 a.m., a more substantial meal at 2:00 p.m., and a light meal again in the early evening. Their diet was composed of vegetables and fruit, raw carrots, cold boiled potatoes, bananas, apples, oranges and bread. Sick animals and young ones were given eggs and milk as necessity demanded. The few flies which they were able to catch were also eaten with apparent relish. Water was furnished at all times by automatic fountains. The visitors must also be reckoned with in an account of the diet of these animals. Thus we should add that the dietary enumerated was supplemented by popcorn, peanuts, chocolate, and so on.

The animals were put at my disposal for a period of three weeks. Through the kindness and coöperative spirit of Dr. Charles V. Noback, veterinary surgeon at the zoo, many of the animals were isolated in separate cages for closer study, and a baby gorilla and baby chimpanzee were made available for testing. Since the period of experimentation was only three weeks, it was impossible to obtain learning curves for the animals. The author contented himself with observing the periods of maximal delay of which the animals were capable and determining differences in capacity by this method.

TABLE 1

Classification of the animals

FAMILY	GENUS	SPECIES	ESTIMATED AGE	SEX
Suborder Anthropoidea				
1. Pongiidae	Pan	Chimpanse	4½ years	Female
2. Pongiidae (gorilla)	Gorilla	Gorilla	4½ years	Female
3. Lasiopygidae (Chacma baboon)	Papio	Porcarius	Adult	Male
4. Lasiopygidae (guinea baboon)	Papio	Papio	Young adult	Female
5. Lasiopygidae (yellow baboon)	Papio	Cynocephalus	Adult	Male
6. Lasiopygidae (drill)	Papio	Leucophaeus	Adult	Male
7. Lasiopygidae (drill)	Papio	Leucophaeus	Baby (2 years)	Male
8. Lasiopygidae (mandrill)	Papio	Sphinx	Adult	Male
9. Lasiopygidae (pig tailed monkey)	Pithecus	Andamanensis	Adult	Male
10. Lasiopygidae (common macaque)	Pithecus	Cinnemalogus	Adult	Male
11. Cebidae (woolly monkey)	Lagothrix	Lagotricha	Young adult	Male
12. Cebidae (spider monkey)	Ateleus	Ater	Adult	Male
13. Cebidae (hooded Capuchin)	Cebus	Unicolor	Adult	Female
14. Cebidae (Capuchin monkey)	Cebus	Capucinus	Old adult	Male
15. Cebidae (Capuchin monkey)	Cebus	Capucinus	Young adult	Male
Suborder Lemuroidea				
16. Nycticibidae (brown ruffed lemur)	Lemur	Variegatus ruber	Adult	Male
17. Nycticibidae (brown mongoose lemur)	Lemur	Rufus	Adult	Female
18. Nycticibidae (gray mongoose lemur)	Lemur	Mongoz	Adult	Male

II. CLASSIFICATION OF THE ANIMALS, THEIR HOUSING AND CARE

The primates tested included the following: 2 anthropoids; 1 baby gorilla and 1 baby chimpanzee; 6 baboons: 1 Chacma baboon, 1 Guinea baboon, 1 Drill (adult), 1 Drill (baby), 1 Mandrill, and 1 yellow baboon; 2 other old world monkeys: 1 pig tailed macaque, 1 common macaque; 5 new world monkeys: 1 woolly monkey, 1 spider monkey, 2 white-faced capuchins, 1 hooded capuchin; 3 lemurs: 1 brown mongoose lemur, 1 gray mongoose lemur and 1 brown ruffed lemur.

These animals are classified in table 1 according to suborder, family, genus and species following the classification of Elliott (1).

The age of these animals in many cases was unknown, the only information available being approximations. Few animals are born in captivity; most of them are supplied by travellers and business firms who know little about their ages. As will be noted, the age of the animal may be an important factor.

All experimentation was carried on between 10:00 a.m. and 2:00 p.m., in the period just before the noon meal. The animals who were tested at the hospital were given no morning meal. These include all the animals but the baboons and the pigtail macaque.

III. APPARATUS

An exact description of the general apparatus used is given in the previous report (2) and need not be repeated in detail. The essential elements were a table (its top flush with floor of cage), painted black, upon which were two white metal cups. The cups were 4 inches in diameter for the larger animals and 1 inch in diameter for the smaller ones. The large cups were inconspicuously attached to the table by wires to keep the position constant. The table represents a slight improvement over that used at at Madison, Wisconsin, in that a sliding top was provided. This made it possible to present the cups to the animals for a choice, smoothly and without disturbance. The relative position of the cups was exactly the same as in the first experiment, their inner edges being 12 inches apart. The experimenter stood directly

behind the table, facing the animal. Only the direct method was used in this experiment.

IV. PROCEDURE

Limitation of time made it necessary to allot only a short time for preliminary training. Each animal was given a thirty minute period of training with the experimental set-up. The method is described by Harlow, Uehling and Maslow (2). In almost all cases this period sufficed and records could be obtained in the succeeding periods of experimentation. Where necessary, as in the case of the lemurs, longer periods of training were given.

As soon as the animal had become accustomed to the general nature of the problem and reacted to it rather than to the experimenter or the environment, tests of delayed reactions were begun. The two cups were placed upon the table top at their specified position just beyond the reach of the animal. A piece of orange was dangled in the air above the cup where it was to go. When the monkey has fixated his eyes upon the orange or reached out after it, it was placed under the cup, the period of delay being timed from the instant the cup covered the food. The experimenter remained motionless in his original position. When the period of delay was up, the table top with the cups on it was slid forward so that the animal could make his choice freely. If he pushed over the right cup, he was allowed to take the food. If he picked the wrong container, he was allowed to see the food but was not allowed to eat it.

The order of presentation was varied to prevent position habits. The time between trials was kept constant as far as possible. Some variation, however, was inevitable, due to distraction, fights with animals in adjoining cages, etc. The periods for delays were 0 seconds, 5, 10, 15, 30, 60 and 120 seconds. If the animal was obviously able to make the easier delays without any trouble, they were hurried through and the longer, more significant, periods studied. Ordinarily, however, the procedure was to give 10 trials of 0 second delay, 10 of 5, etc. . . . . The period of experiment on an animal was terminated for the day when he failed definitely on a delay period. Failure was usually indicated by a

wandering of attention in the case of the milder animals and active anger in the case of the more vicious individuals. In only 1 case was the delay period taken up to 180 seconds (with Tommy, the yellow baboon). This procedure was always followed, since we have shown in the previous paper (2) that putting long delays at the end of the experimental period in no way prejudices the results.

Work was carried on from 10:00 a.m. to 2:00 p.m. daily, as many animals being tested in this time as was possible. The animals chosen for experimentation on any given day were selected because of their calmness at the moment and high state of motivation. If an animal did not look promising, work on him was stopped immediately and another subject begun. Thus the data that we present are not average, but favorable, representations of ability. Also, the results were obtained in no set number of days, but were obtained in random order.

V. RESULTS

Comparison of genera

We shall adopt as a basis for comparison a grouping of primates into (*a*) anthropoids, (*b*) old world (Catarrhine) monkeys, which include baboons, macaques, guenons, and mangabeys, (*c*) new world (Platyrrhine) monkeys, and (*d*) lemurs. The baboons have usually been considered the most primitive of the old world monkeys because of their dog-like build. Our results, however, show that the baboons are definitely superior, rather than inferior, to the rest of the Catarrhine monkeys we have tested. At the Bronx Park zoo, 6 baboons were tested: 5 adults, and 1 young drill. At the Madison zoo, 4 adults and 2 young were tested. A cursory inspection of their performance and a comparison with the other 13 Catarrhine monkeys at Madison and the one at Bronx Park demonstrate unquestionably a definite superiority in favor of the baboons. In only 1 case did this superiority apparently fail. The Mandrill at the Bronx zoo shows a steady falling off until at the 60 second delay his score is only a little better than chance. This may probably be explained by the fact that this Mandrill was an extremely ill-tempered animal and never became

reconciled to the experimental situation. Many of his responses were toward the cups rather than toward the food beneath them. The Mandrill at the Madison zoo did much better, being one of our best experimental animals.

We could not try the Chacma baboon on the 120 second delay because of press of time, but he demonstrated a clear superiority when tried on the 60 second delay, getting 85 per cent of his choices correct. We are probably justified in extrapolating and assuming that he could pass the criterion on the 120 second delay.

We might add that a 120 second response is typically a baboon response, while the other old world monkeys typically can do, at best, from 30 to 60 second delays.

It is unfortunate that our results on anthropoids are numerically so inadequate. The one adult tested at Madison before his death demonstrated an unquestionable ability. We have records of his success in 600 second delays. The two anthropoids available at Bronx zoo were both little more than babies, about four and one-half years of age. One was a gorilla, the other a chimpanzee. The latter was a lively and extremely mischievous animal, while the former was very lazy and unmotivated. Furthermore, these were the only animals in the zoo that were fed well in the morning. Their motivation was never at a very high level, and the only results obtained were gotten by patiently coaxing them with bits of chocolate and raisins. The impression they made upon the experimenter was one of undoubted ability but poor motivation. Some degree of hunger is absolutely necessary in order that the experiment may go on. With well-fed animals like these we could hope for little.

We were fortunate in being able to test 5 South American (Platyrrhine) monkeys, inasmuch as we had tested previously at Madison only two very timid Cebus monkeys. The results obtained at Bronx zoo establish more firmly the fact that the South American monkeys were distinctly inferior to Old World monkeys as tested by the delayed reaction. The utmost that any of the South American animals were capable of was 15 second delays. This result was especially striking because as experimental animals the Woolly monkey and older white-faced capu-

chin were ideal. They were always hungry and were very willing. They were not timid, and were perfectly tame. We obtained here a convincing demonstration of their innate limitations.

The lemurs were unquestionably the poorest performers. Without exception, it took them a long time before they could even grasp the problem, and when they did so, their performances were poor. They were capable of only 5 second delays at most. The Lemur Variegatus at Madison after many months of urging and coaxing suddenly displayed ability enough to master the 15 second delay. It is possible that the lemurs we tested at Bronx zoo would eventually have done the same.

Harlow on the basis of the results obtained at the Madison zoo with 1 lemur and 2 cebus monkeys, has tentatively grouped lemurs and Platyrrhine monkeys together. This is in accordance with Tilney's suggested division of primates into four groups: (1) man, (2) higher anthropoids (gorilla, chimpanzee and orang-outan), (3) intermediate primates, including gibbon and all the Cercopithecidae, and (4) the lemurs and Platyrrhine monkeys.

It is now apparent, on the basis of the results here presented, that there is a real difference, qualitative as well as quantitative, between the two groups. The possible fact that the Bronx Park lemur would eventually have come up to the Platyrrhine level of performance does not argue away the very real difference observable in their reaction to the problem. The lemur attains this level, if he does attain it, by travail and labor, and after long training. The South American monkey attacks the problem and conquers it, usually in a day or two. So far as their reaction to the experiment is concerned, they are different animals.

Bodily orientation

The problem of determining the rôle that bodily orientation plays in delayed reaction has already been discussed by Tinklepaugh (4) and Harlow, Uehling and Maslow (2). The latter conducted a brief experiment, showing that in the better animals it plays no part even though it usually plays some rôle in the responses of less able animals. The baboon typically roams about during the period of delay. I have seen the yellow baboon

during a period of long delay go into his indoor compartment and stay there until he was called. He then came back and made the correct response without any hesitation. This was during an 180 second delay. Harlow (3) reported that during the delay the animal being tested had copulated with his mate and then had come back to make the correct response.

In the case of the lemurs and Platyrrhine monkeys a different situation obtains. Harlow found that unfavorable orientation made little difference in the correct responses of the four baboons he tested, but that such orientation was of some importance in the case of the gibbon; and that in all the other animals tested, presentation of the correct cup in an unfavorable position reduced the number of right responses materially, in 1 case almost abolishing them. This factor of course may be disregarded in the interpretation of the present data, since the cups were presented neither favorably nor unfavorably, but in a neutral position.

Relationship of age to ability

No general conclusions can be based on the small number of age comparisons possible either in this experiment or the previous one. At the Madison zoo, 2 young mandrills were far inferior to the mature adults. At Bronx zoo, a similar situation obtained. A young Drill, perhaps two years old, was able definitely to complete 15 second delays, making correct responses in 62 per cent of the cases. When delayed for 30 seconds, he responded correctly only 55 per cent of the time, which is but little better than chance. This is much inferior to the mature Drill, who did as well with 120 second delays.

The author would not feel justified in drawing any conclusions on the basis of the results from the young gorilla and the young chimpanzee. If we should interpret the data as they stand, it would of course be necessary to say that they were inferior to the adult animals of Yerkes (5), (6) and others. However, the complications introduced by their playfulness and their lack of motivation necessitate extreme caution with respect to conclusions.

Another age comparison was furnished us by the two white-faced Capuchins. Both were mature but one was definitely an

old animal. He was large for a Cebus monkey, and the keepers were certain that he was old, although he was just as lively and well-motivated as the other. Table 2 shows that his performance is superior to that of the younger animal, who was unsuccessful beyond 10 second delays. The older one was successful up to 15 seconds.

TABLE 2

Summary of correct responses

ANIMAL	PERIOD OF DELAY (SECONDS)															
	0		5		10		15		30		60		120		180	
	N	%C	N	%C	N	%C	N	%C	N	%C	N	%C	N	%C	N	%C
Young chimpanzee	30	93	30	83			50	76	48	61	28	82	25	60		
Young gorilla	20	85	40	83			39	74	36	72	31	52	1	100		
Chacma baboon	5	100	15	93			20	85	20	85	20	85				
Hamadryas baboon	5	100	5	100			5	100	5	100	10	70	10	60		
Drill	5	100	20	95			31	84	35	90	35	77	17	60	2	0
Mandrill	5	100	10	100			15	66	20	65	7	57				
Yellow baboon	5	100	20	95			15	87	20	80	30	83	22	68	10	70
Baby mandrill	44	81	40	75			37	62	40	55	20	70	7	0		
Pig tailed monkey	5	100	15	93			24	75	25	76	10	60				
Common macaque	Too ferocious to attend to experimental situation															
Woolly monkey	50	88	40	85			49	82	52	45	20	55				
Spider monkey	Too sick to respond. Died soon after															
Hooded Capuchin	15	93	13	77	10	50										
White faced Capuchin	28	96	15	60	19	74	25	72	20	45						
White faced Capuchin (young)	34	85	30	77	10	70	10	20								
Brown ruffed lemur	30	80	20	75												
Gray mongoose lemur	21	67	8	75												
Brown mongoose lemur	Too shy to respond															

N, number of trials. %C, percentage of trials correct.

For further comparisons with other investigations, the reader is referred to the article by Harlow, Uehling and Maslow (2). Comparisons of our results with data there presented have been drawn throughout this paper.

VI. SUMMARY AND CONCLUSIONS

1. Tests of delayed reaction were carried out at the Bronx Park zoo in New York City on 18 primates. The results ob-

tained indicate increased capacity in order from the lemur, Platyrrhine monkeys, Catarrhine monkeys to the baboons. Attention is especially called to the clear superiority of the latter, since they have often been thought inferior.

2. There was found, in these tests, a consistent performance characteristic of the groups.

3. Some indications of age differences in primate capacity for delay are cited.

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